

Exploratory Data Analysis and Machine Learning for E-Commerce Sales Performance Optimization

Daniel Satrio Pamungkas
Computer Science Department
School of Computer Science, Bina Nusantara University
Jakarta, Indonesia 11480
daniel.pamungkas@binus.ac.id

Deken Febriano Putra
Computer Science Department
School of Computer Science, Bina Nusantara University
Jakarta, Indonesia 11480
deken.putra@binus.ac.id

Maria Maureen Linggar
Computer Science Department
School of Computer Science, Bina Nusantara University
Jakarta, Indonesia 11480
maria.linggar@binus.ac.id

Eko Setyo Purwanto
Computer Science Department
School of Computer Science, Bina Nusantara University
Jakarta, Indonesia 11480
eko.purwanto@binus.ac.id

Abstract—The Industrial growth in E-Commerce has produced a large volume worth of data that will continue to grow in the near future. This research aims to further analyze the business performance in the E-Commerce district to fully utilize Exploratory Data Analysis and machine learning for the Amazon Sales Report dataset. Preprocessing methods include data cleaning, feature engineering, data visualization, and model training. An exploratory data analysis was done to further identify sales patterns, consumer behaviour, and non-linear relationships between attributes. The result shows that the product category of “Set” and “Kurta” shows high sales. Product sizes of M, L, and XL have also dominated in sales transactions. The trend analysis has also shown an increase in daily sales from April to June 2022. The research is divided into 3 major classifications, those are: Random Forest, Decision Tree, and Logistic Regression to predict “Shipped” and “Cancelled” order status. Every model has shown an accurate prediction up to 95%. Random Forest showed a great performance based on precision, recall, f1-score, and ROC-AUC. The result shows that combining Exploratory Data Analysis and machine learning can help companies to increase their marketing strategy, operational efficiency, and decision-making capabilities in the E-Commerce district.

Keywords—*Exploratory Data Analysis, Machine Learning, E-Commerce, Random Forest, Decision Tree, Logistic Regression*

INTRODUCTION

The Industrial growth for E-Commerce has successfully produced a massive volume of transactional data that will keep on growing in the near future. This data includes all kinds of information, starting with selling information, product categories, region, delivery method, and much more. This calls for the need to use the data optimally so companies may start making business decisions far more effectively than before[1]. One of the many approaches to this attempt is Exploratory Data Analysis, which is an early process to identify patterns, connections, trends, and anomalies from the dataset before researchers may start analysing even further or before using machine learning [2]. The past research has mentioned that Exploratory Data Analysis and machine learning could further help the business sector in increasing its industrial capacity by enhancing its operational efficiency, marketing strategy, and consumer behaviours[3][4][5].

Further evaluating the business as a whole requires a dataset. The E-Commerce sales data from Kaggle offers many kinds of attributes varying from sales, quantity, discount, profit, and many more. This dataset can be fully used to evaluate the business performance as a whole [3]. With the usage of Exploratory Data Analysis, companies may identify which product category offers the best or the worst profit, what's the best-selling performance, and the connections of discount towards companies' profits [6][7][8]. Visualization of the data may also point out the patterns of consumer behaviours and transactional trends in a much clearer manner. With these capabilities, companies can prepare a marketing strategy to be better

suited to what the current market demands [9][10][11][12].

With Exploratory Data Analysis, this research intends to analyse the E-Commerce dataset provided by Kaggle to optimize marketing strategies and companies' profit. Analysis is required to use data preprocessing, descriptive statistics, and visualization techniques to give a much clearer clarity for better insights that are much more relevant for business performances [13][14][15][16]. Researchers hoped that the outcome of this research could help companies to better understand what influenced the profit and sales. Researchers also hoped that they could improve their business strategy efficiency in the future with our results. With those results aside, researchers also hoped that this research could potentially be a strong reference for further enhancements on business analytics systems in the E-Commerce sector and digital retail [17][18][19].

LITERATURE REVIEW

2.1. E-Commerce Growth and Digital Commerce

The development of digital technology and internet access is growing very fast. The development of digital technology has sped up the growth of the E-Commerce industry across the world. E-Commerce allows businesses and consumers to do transactions electronically through digital platforms. The use of digital platforms makes the shopping processes more efficient and more accessible. The rise of smartphone usage also supports the development of E-Commerce. Online payment systems help simplify digital transactions. The digital marketplace also plays a major role in expanding the digital commerce activities across various industries [6][20].

Along with the growth of E-commerce that is continuously increasing, companies produce a lot of transaction data every day. The transaction data consists of the customer purchase history, product categories, sales revenue, shipping information, customer preferences, and transaction status. The continuous growth of transaction data always creates opportunities along with challenges for businesses. The company has to manage and analyze the data in large volumes effectively. This is needed so that the company remains competitive in the digital market environment [10][17].

The rise of digital commerce has changed customer behavior and market competition. The customers these days hope and expect that they have services that are faster. The customers these days also want recommendations that are personalized. Other than that, the customers these days also hope and expect to have a better shopping experience. Because of this, the businesses have to always adjust their strategies to meet the customer's needs. Businesses need to use the analytical approach and the data-driven technology. The

purpose of this is to better understand the market needs and the preferences of the customer [11][15][21].

2.2. Transactional Data in E-Commerce

The transaction data is one of the most important and valuable assets in e-commerce businesses. The transaction data consists of detailed information about the customer activities and the business operations. Every online transaction produces data, such as order quantity, product categories, payment information, shipping methods, sales volume, and order status. This data can be analysed to know and identify the customer purchasing patterns, the sales performance, the customer preferences, and the business operational efficiency [8][12].

E-commerce platforms generate a lot of transactional data, this large volume of data can be useful for businesses' comprehensive analysis. Through data processing, businesses can identify plenty of analysis results, for example, customer purchasing trends, locate regions with the highest profits, products with the highest gross, and also identify seasonal demand patterns. By using this transactional data, it can help businesses can evaluate their performance and improve their strategic planning processes [16][18].

However, if businesses use raw data to make an analysis, it can result in a misleading analysis. Because raw transactional data often contains some missing values, inconsistent data, and formatting issues. Due to this, data preprocessing, such as data cleaning, is crucial so that the analysis results can be more effective and accurate rather than using raw data [19].

2.3. Data-Driven Decision Making

Business decisions that are based on data analysis mean that the company really depends on analytical insights to support business decision-making, so that the company does not only use intuition to make future actions. E-commerce itself highly relies on this method, and this method is really important because E-commerce has a rapidly changing environment and many competitors in the market. For this reason, if businesses use analytical insights effectively, they can bring improvements to operational efficiency, profitability, and customer satisfaction [1][2].

Through data analysis, it is easier for companies to study customer behaviour, evaluate marketing performance, and observe sales trends. Through the analytical insights, businesses will understand numerous findings such as which products perform well, which regions have the highest sales, and which marketing strategies can lead businesses to better outcomes. These findings make companies develop more effective strategies and improve business performance [21].

Business intelligence systems and analytical tools are commonly used as the foundation for E-commerce. They help support the decision-making processes. These technologies help companies convert raw data into better analytical insights and useful visualizations to make better, faster, and more accurate decisions to follow market trends [22].

2.4. The Role of Analytics in Business Performance

Improving e-commerce performance is heavily reliant on data analytics. By using data analytics methods such as Exploratory Data Analysis (EDA), machine learning, and statistical analysis, a company's understanding of business patterns, trends, and anomalies can be improved, and it also helps to discover problems and new business opportunities [23].

The main use of data analytics in E-commerce is to identify sales trends. The sales data analysis that is conducted periodically helps the business to identify the peak sales period. Data analytics can be used to detect seasonal trends in the data. Data analysis helps understand the customer's purchasing behavior. The product category analysis helps companies to identify products with the highest sales. By using data analytics, companies can also identify which product makes the highest profits. The geographical analysis helps evaluate sales performance across various regions. Data analytics also helps

understand the sales performance based on different customer categories [24].

Data visualization helps improve the data analysis process. The graphs, the charts, and the dashboard help present the data more clearly. Visualization makes it easy to understand the patterns in the data and the complex information in the data. Data visualization is very useful in business decision-making. Machine learning methods, such as classification and prediction models, help perform data prediction. Machine learning can be used to predict customer behavior. Machine learning can also be used to predict transaction results [25].

METHODOLOGY

3.1. Research Approach

This research uses a quantitative analysis approach. The method that is used is Exploratory Data Analysis (EDA) and machine learning. The data that is analysed is E-commerce transaction data from Kaggle, which is named the Amazon Sale Report dataset. The focus of this research consists of sales trends, the customer purchasing behaviour, product performance, and order status patterns. The analysis methods that are used are descriptive statistics, data visualization, aggregation analysis, and classification modelling. The purpose of this research is to produce useful business insights or business information from the data.

3.2. Workflow

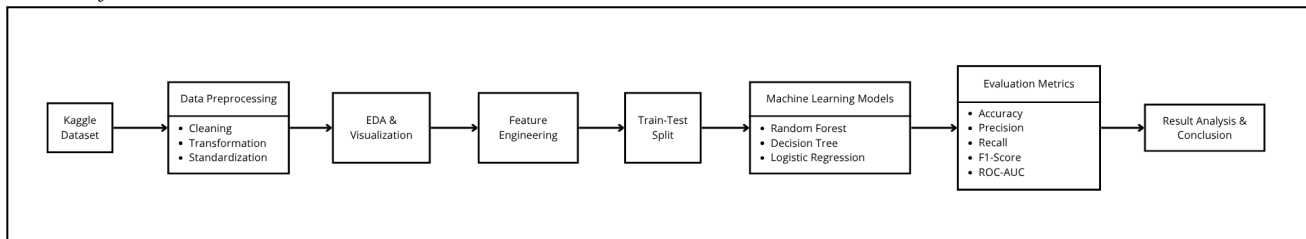


Fig 1. Research Workflow

3.3. Dataset

This research uses the e-commerce sales dataset (Amazon Sale Report dataset) that is obtained from Kaggle. The e-commerce sales dataset (Amazon Sale Report dataset) consists of transactional data and financial information that is about the online sales activities. The data consists of sales channels and order fulfilment methods. There are product attributes in the dataset, such as SKU, product category, and size. There are also pricing variables and transactional variables, such as sales amount, quantity, fulfilment method, sales channels, customer payment method, and order status. These variables are used to analyse the sales performance, the customer behaviour, the operational efficiency, and the

profitability. The e-commerce sales dataset is suitable to be used for Exploratory Data Analysis (EDA) and for the application of machine learning in e-commerce.

3.4. Data Preprocessing

3.4.1. Data Cleaning

The data cleaning process is conducted to handle and address various data quality problems. The data duplicates are identified based on Order ID, but in this stage, the data duplicates are only detected and not yet removed from the dataset. The column that is irrelevant, such as Unnamed: 22, is removed because the column does not give important information for analysis. The missing

values in the dataset are handled differently based on the variable type. For the categorical data, such as Courier Status, currency, promotion-ids, and the fulfilled-by, the missing values are filled with default values that are appropriate so that the data remains consistent. The data that have missing values on the important columns, such as ship-city, ship-state, ship-country, and ship-postal-code, are removed from the dataset to maintain the accuracy and the reliability of the data. The final purpose of this data cleaning process is to ensure that the dataset is cleaner than before this process was conducted and to ensure that the dataset is ready to be used for the next analysis.

3.4.2. Data Transformation

Several columns have data types that are not suitable for the analytical requirements, so data type transformation is performed. The Date column is converted into a datetime format to support time-based analysis. In addition, ship-postal-code is converted into a string type to prevent numerical misinterpretation. These transformations are essential to ensure data consistency before performing aggregation or visualization.

3.4.3. Data Standardization

The data standardization process is conducted on the ship-state column to improve the consistency of the categorical data. The text format is normalized by converting all of the letters to uppercase and removing the extra spaces. The writing variation and typographical errors in the state name are corrected and standardized into a single uniform format. The purpose of this data standardization process is to ensure that every state has its own representation that is consistent in the dataset. With data that is already uniform and standardized, the results of the geographical analysis will become more accurate and reliable.

3.4.4. Feature Engineering

The feature engineering stage is performed by creating new variables derived from the Date column, namely Year, Month, and Day. These variables are used to support sales trend analysis over time. In addition, a new target variable called Status_Simple is created by simplifying the order status into two main categories, namely “Shipped” and “Cancelled”, to facilitate machine learning modeling.

3.5. Exploratory Data Analysis

The purpose of conducting the Exploratory Data Analysis (EDA) is to understand the structure, the distribution, and the patterns in the e-commerce sales dataset (Amazon Sale Report) after the data preprocessing

process. The data preprocessing process that is conducted before consists of handling the missing values, removing the data duplicates, and standardizing the categorical variables. The analysis that is conducted consists of descriptive statistics that are used to summarize the data. The data quality inspection is conducted to ensure that the data is already clean and ready to be analyzed. The temporal feature extraction is conducted to see and understand the pattern that is based on time. The univariate analysis and the multivariate analysis are used to understand the variables individually and also to understand the relationships between variables. The purpose of this analysis is to identify the sales trend, the customer purchase behavior, and the product performance. The visualization techniques, such as line charts and bar plots, are used to simplify the understanding and to produce business insight that is clearer. The results of this EDA are then used as a foundation for the next stage, which is the predictive modelling stage.

3.6. Data Visualization

The purpose of the data visualization process is to transform the transactional data into information that is more understandable visually. Before visualization, the data has already gone through the data preprocessing stage and the feature engineering stage. The dataset is ensured to have data types that are consistent and have the appropriate category representation. The aggregation process is conducted to summarize the important business metrics, such as the sales amount and the number of orders. The analysis is conducted based on several dimensions such as temporal, product category, and product size. Several visualization techniques that are used are line charts, bar charts, and distribution plots. The purpose of these visualizations is to identify and see the sales trends, to identify seasonal trends, and to analyze the product performance. The results of the visualization help give an interpretation that is clearer for business decision-making.

3.7. Model Training

The model training process is conducted after the data cleaning stage, the preprocessing stage, and the Exploratory Data Analysis (EDA) stage. The features that are used for prediction include Category, Size, Amount, and Qty. The target that is going to be predicted by the machine learning models is the order status that is simplified into two classes: “Shipped” and “Cancelled”. Label Encoding is used to transform the categorical variable into numerical form. The data rows that have missing values are removed. This is done to maintain data consistency. The data is then divided into training data and testing data with a ratio of 70:30. Three classification

algorithms are used to compare the performance. The three classification algorithms are the Random Forest algorithm, the Decision Tree algorithm, and the Logistic Regression algorithm. All of the models are trained using the same dataset so that the evaluation results of these models can be compared fairly. The Random Forest model is configured with 100 decision trees to improve the stability of the predictions and to reduce the overfitting risk.

3.8. Evaluation Approach

The evaluation is conducted to measure the prediction performance from the classification models that are used in this research. The evaluation methods that are used in this research include confusion matrix, accuracy, precision, recall, F1-score, and ROC-AUC score. These evaluation metrics are used to measure the effectiveness of the models in classifying the order status into the “Shipped” class and the “Cancelled” class. The use of the precision metrics, the recall metrics, and the F1-score metrics is important because the dataset has a class imbalance condition, where the number of “Shipped” data is higher compared to the “Cancelled” data. The ROC-AUC score is used to measure the ability of the models to differentiate the two classes across various threshold values. The results of the evaluation of the Random Forest model, the Decision Tree model, and the Logistic Regression model are compared to determine which model is the most accurate, stable, and robust in predicting the e-commerce order status.

RESULTS & DISCUSSION

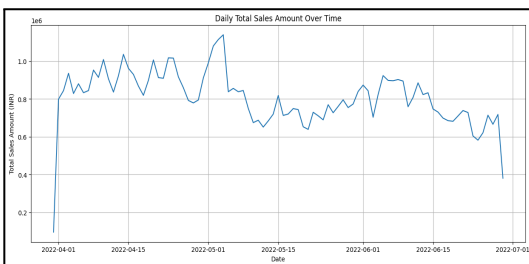


Fig 2. Daily Total Sales Amount Over Time

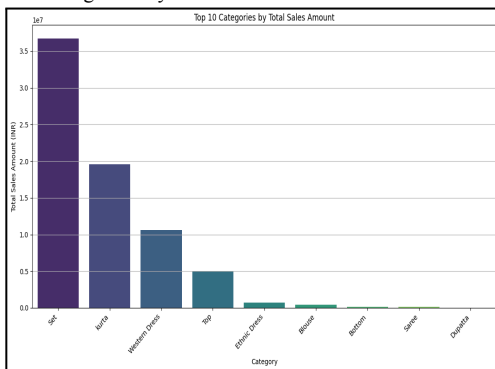


Fig 3. Top 10 Categories by Total Sales Amount

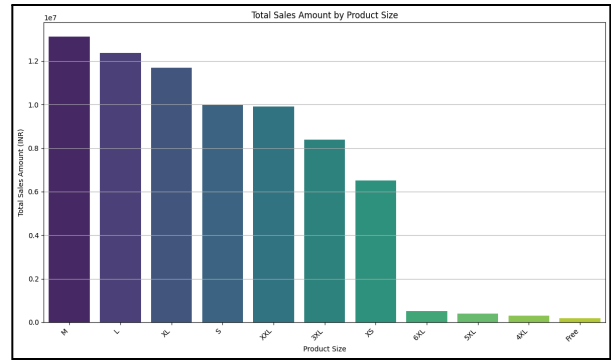


Fig 4. Total Sales Amount by Product Size

The time-series graph shows the daily total sales amount that started from April 2022 to June 2022. Based on the time-series graph, total sales on the e-commerce platform experienced changes over time. These total sales changes show that there is a customer purchase trend that is different every day. This time-series graph can help in understanding the sales patterns and understanding the customer behavior during a certain period. At the beginning of April, total sales increased significantly and then stayed stable at 0.8 to 1 million INR. The peak of the total sales amount was in early May, but it lasted for only a short time. Then, it moderately decreased from 1.1 million INR to 0.7 million INR in the middle of May, which may be caused by seasonal trends, lower customer demand, or other factors, and it continued to vary between 0.6 and 0.9 million INR.

The bar chart visualization shows that the “Set” product category produced the highest total sales. The “Kurta” product category, the “Western Dress” product category, and the “Top” product category also have high total sales following the “Set” product category. Meanwhile, the product categories, such as “Ethnic Dress”, “Blouse”, and “Dupatta”, have total sales that are relatively lower. These results show that customers tend to purchase more products from certain categories that are more popular. The product category analysis is important to be conducted to help optimize the inventory stock and to help optimize the marketing strategy on e-commerce platforms.

The bar chart shows that the product sizes, such as Medium (M) product size, Large (L) product size, and extra-large (XL) product size, produce the highest total sales amounts. This shows that the standard product sizes give the largest contribution to the sales revenue in the E-Commerce dataset. The oversized product size, such as 3XL-6XL, and the Free category product size have sales amounts that are relatively lower. The low sales on these product sizes show that the customer demand for the oversized product size and the Free category product size tends to be lower. The customer's purchase behavior is more focused on the products compared to the standard size products. This finding can help businesses optimize

their inventory management and product allocation. The results of the analysis can also be used to build and develop marketing strategies that are more effective to improve the sales performance overall.

Random Forest Classifier Model

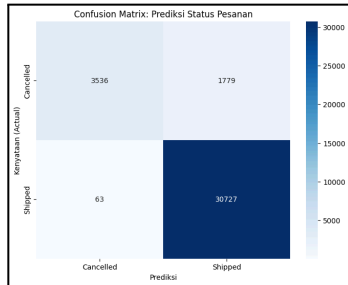


Fig 5. Confusion Matrix of The Random Forest Classifier Model

Decision Tree Classifier Model

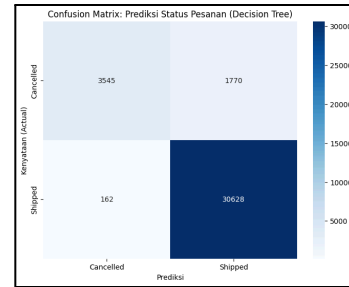


Fig 6. Confusion Matrix of The Decision Tree Classifier Model

Logistic Regression Model

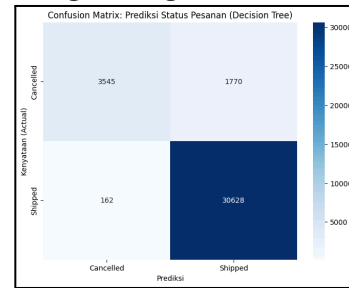


Fig 7. Confusion Matrix of The Logistic Regression Model

Table 1. Performance Comparison of Machine Learning Classification Models

	Random Forest Classifier Model				Decision Tree Classifier Model				Logistic Regression Classifier Model			
	precision	recall	f1-score	support	precision	recall	f1-score	support	precision	recall	f1-score	support
Cancelled	0.98	0.67	0.79	5315	0.96	0.67	0.79	5315	1.00	0.66	0.80	5315
Shipped	0.95	1.00	0.97	30790	0.95	0.99	0.97	30790	0.94	1.00	0.97	30790
accuracy	-	-	0.95	36105	-	-	0.95	36105	-	-	0.95	36105
macro avg	0.96	0.83	0.88	36105	0.95	0.83	0.88	36105	0.97	0.83	0.88	36105
weighted avg	0.95	0.95	0.94	36105	0.95	0.95	0.94	36105	0.95	0.95	0.95	36105
ROC AUC Score	0.85				0.85				0.84			

This research compares the performance of three classification models. The three classification models are the Random Forest model, the Decision Tree model, and the Logistic Regression model. The purpose of this research is to predict the order status category that is simplified to be “Shipped” or “Cancelled”. The model evaluation process uses the Confusion Matrix, Precision, Recall, F1-Score, Accuracy, Macro Average, Weighted Average, and ROC-AUC Score. The reason for using various important evaluation metrics is that the dataset experiences class imbalance. The reason for the class imbalance condition in the dataset is that the amount of the “Shipped” data is more than the amount of the “Cancelled” data. By using these evaluation metrics, this research can identify and understand which model is the best in classifying the order status categories.

Based on the results of the evaluation, all of the three machine learning models have accuracy scores that are almost similar. The accuracy scores of the three machine

learning models are approximately 95%. This shows that all of the three machine learning models can make order status predictions effectively. The high accuracy scores show that most of the data was successfully classified by the three machine learning models. However, accuracy alone was not enough to evaluate the overall performance of the three machine learning models. This was because the dataset was dominated by the order data that had the “Shipped” status compared to the “Cancelled” status. This condition could cause the accuracy scores to seem high, although the machine learning models might not be able to predict all of the classes effectively. This is why further analysis is needed using precision, recall, and F1-score to compare the performance of the three machine learning models effectively and in more detail.

Based on Table 1, the Random Forest Classifier produced a strong classification performance with a precision value of 0.98 for the “Cancelled” class and 0.95 for the “Shipped” class. The recall score for the

“Cancelled” class reached 0.67, while the “Shipped” class achieved a perfect recall score of 1.00. This indicates that Random Forest was highly effective in identifying shipped orders with very few false negatives. The F1-score values of 0.79 for “Cancelled” and 0.97 for “Shipped” also demonstrated balanced classification performance. In addition, the macro average F1-score of 0.88 showed that the model maintained good predictive capability across both classes despite the imbalance in the dataset. From the confusion matrix analysis, as shown in Fig 5, Random Forest generated relatively fewer misclassifications compared to the other models, especially in predicting cancelled orders.

According to Table 2, the Decision Tree Classifier model that was used in this research produced an accuracy score that is 95%. But the classification performance of the Decision Tree Classifier model was slightly lower than that of the Random Forest model. The Decision Tree Classifier model produces a precision score for the “Cancelled” class that is 0.96. Meanwhile, the recall score for the “Cancelled” class remained at 0.67, similar to the Random Forest model. For the “Shipped” class, the Decision Tree Classifier model produces a precision score of 0.95 and a recall score of 0.99. Although the Decision Tree Classifier model showed relatively competitive performance, the F1-score value for the “Cancelled” class was only 0.79. This score was slightly lower compared to the Random Forest model in terms of classification stability. Based on Fig 6, the Decision Tree Classifier model produced more false positive and false negative predictions. This shows that the model was more sensitive to overfitting. Other than that, the Decision Tree Classifier model was also less stable when handling the testing data that had never been seen before.

Table 3 shows that the Logistic Regression model showed different classification characteristics compared to the tree-based models, such as the Decision Tree model and the Random Forest model. This model produces the highest precision score for the “Cancelled” class, that is reaching 1.00. This shows that almost all of the orders that are predicted as “Cancelled” were actually cancelled orders. However, the recall score for the “Cancelled” class was only 0.66. This recall score shows that the Logistic Regression model still failed in detecting several orders that are actually cancelled. For the “Shipped” class, the Logistic Regression model produces a recall score of 1.00 and an F1-score of 0.97. These results show that the Logistic Regression model has excellent performance in detecting orders that are successfully shipped. The weighted average F1-score of 0.95 was slightly higher compared to the other models. This shows that the Logistic Regression model gives consistent overall performance. Based on Fig 7, the Logistic Regression model tended to prioritize prediction confidence over the balance of the detection between classes. The results are

that the Logistic Regression model produces a recall score that is lower for the “Cancelled” class.

Based on the evaluation results shown in Tables 1–3 and Figs. 5–7, the Random Forest Classifier model can be assumed as the model with the best performance compared to the three algorithms that are used in this research. This is because the Random Forest model gives performance that is the most balanced across the precision metrics, the recall metrics, and the F1-score metrics. The Random Forest model was also more robust in handling class imbalance. Other than that, the Random Forest model produces generalization ability that is better compared to the Decision Tree model and the Logistic Regression model. The Decision Tree model has interpretations that are simpler and easier to understand, but the Decision Tree model shows stability in the performance that is slightly weaker. Meanwhile, the Logistic Regression model has a precision score that is excellent, but the Logistic Regression model has a recall score that is lower for the cancelled orders class. These research results show that the ensemble-based approaches, such as Random Forest, are more suitable to be used to predict E-Commerce order status. This is because the Random Forest model can better capture complex relationships between transaction features and order status results and can minimize prediction errors.

CONCLUSION

This research demonstrates that the combination of Exploratory Data Analysis (EDA) and machine learning techniques can effectively support the optimization of E-Commerce sales performance using the Amazon Sales Report dataset. The EDA process successfully identified important business patterns related to sales trends, customer purchasing behavior, product categories, and product sizes. The findings revealed that categories such as “Set” and “Kurta” generated the highest sales revenue, while standard clothing sizes such as M, L, and XL dominated customer purchases. Furthermore, the implementation of Random Forest, Decision Tree, and Logistic Regression classification models showed that machine learning can accurately predict simplified order status categories with approximately 95% accuracy. Among the evaluated models, Random Forest demonstrated the most balanced and robust performance in handling class imbalance and minimizing prediction errors. Overall, the results indicate that integrating EDA and machine learning provides valuable analytical insights and predictive capabilities that can assist businesses in improving inventory management, operational efficiency, marketing strategies, and data-driven decision-making within the E-Commerce industry.

REFERENCES

- [1] Saeed, S. E., Hammad, M., & Alqaddoumi, A. (2022). Predicting customer's subscription response to bank telemarketing campaign based on machine learning algorithms. In 2022 International Conference on Decision Aid Sciences and Applications (DASA) (pp. xxx-xxx). IEEE. <https://doi.org/10.1109/DASA54658.2022.9765152>
- [2] Nyasala, U. S., Lingannagari, N. R., Kakarla, P., Gujjala, U. K., & Neeruganti, S. H. V. (2025). Predictive analytics with machine learning for fraud detection of online marketing transactions. In AIP Conference Proceedings (Vol. 3237, Issue 1, Article 020029). AIP Publishing. <https://doi.org/10.1063/5.0248862>
- [3] Minatogawa, V. L. F., Franco, M. M. V., Rampasso, I. S., Anholon, R., Quadros, R., Durán, O., & Batocchio, A. (2020). Operationalizing business model innovation through big data analytics for sustainable organizations. *Sustainability*, 12(1), 277. <https://doi.org/10.3390/su12010277>
- [4] Sarker, I. H. (2021). Machine learning: Algorithms, real-world applications and research directions. *SN Computer Science*, 2(3). <https://doi.org/10.1007/s42979-021-00592-x>
- [5] Dou, X. (2020). Online purchase behavior prediction and analysis using ensemble learning. In 2020 IEEE 5th International Conference on Cloud Computing and Big Data Analytics (ICCCBDA) (pp. 532-536). IEEE. <https://doi.org/10.1109/ICCCBDA49378.2020.9095554>
- [6] Yemunarane, K., Chandramowleeswaran, G., Subramani, K., Alkhayyat, A., & Srinivas, G. (2024). Development and management of e-commerce information systems using edge computing and neural networks. *Indian Journal of Information Sources and Services*, 14(2), 153-159. <https://doi.org/10.51983/ijiss-2024.14.2.22>
- [7] Liao, M., Fang, J., Han, L., Wen, L., Zheng, Q., & Xia, G. (2023). Boosting eCommerce sales with livestreaming in B2B marketplace: A perspective on live streamers' competencies. *Journal of Business Research*, 167, 114167. <https://doi.org/10.1016/j.jbusres.2023.114167>
- [8] Nagpal, G., Bishnoi, G. K., Dharmi, H. S., & Vijayvargia, A. (2021). Use of data analytics to increase the efficiency of last mile logistics for ecommerce deliveries. In B. Patil & M. Vohra (Eds.), *Handbook of research on engineering, business, and healthcare applications of data science and analytics* (pp. 167-180). IGI Global. <https://doi.org/10.4018/978-1-7998-3053-5.ch009>
- [9] Shreenidhi, A., Ruba Roshini S., & Ranjana, P. (2024). Walmart sales forecasting using time series analysis. In 2024 International Conference on Advances in Data Engineering and Intelligent Computing Systems (ADICS 2024). IEEE. <https://doi.org/10.1109/ADICS58448.2024.10533547>
- [10] Stoica, F., & Pelican, E. (2025). A machine learning approach of enhancing eCommerce solutions. *Expert Systems with Applications*, 274, 126997. <https://doi.org/10.1016/j.eswa.2025.126997>
- [11] Afzal, M., Rahman, S., Singh, D., & Imran, A. (2024). Cross-sector application of machine learning in telecommunications: Enhancing customer retention through comparative analysis of ensemble methods. *IEEE Access*, 12, 115256-115267. <https://doi.org/10.1109/ACCESS.2024.3445281>
- [12] Jahan, I., & Farah Sanam, T. (2022). An improved machine learning based customer churn prediction for insight and recommendation in e-commerce. In *Proceedings of the 2022 25th International Conference on Computer and Information Technology (ICCIT)* (pp. 1-6). IEEE. <https://doi.org/10.1109/ICCIT57492.2022.10054771>
- [13] Wu, S., Yau, W.-C., Ong, T.-S., & Chong, S.-C. (2021). Integrated churn prediction and customer segmentation framework for telco business. *IEEE Access*, 9, 62118-62136. <https://doi.org/10.1109/ACCESS.2021.3073776>
- [14] Zhang, M., & Wu, L. (2025). Sales forecasting and data-driven marketing strategies for e-commerce platforms using XGBoost. *International Journal of Intelligent Information Technologies*, 21(1). <https://doi.org/10.4018/IJIIT.382563>
- [15] Lo, H.-C., Chang, W.-J., & Chen, I.-H. (2026). From purchase to return: How personalized E-commerce recommendations shape consumer behavior. *Journal of Retailing and Consumer Services*, 88, 104459. <https://doi.org/10.1016/j.jretconser.2025.104459>
- [16] Bhavana, E., Vyshnavi, A., Saxena, S., Sreeja, B. S. L., & Druthi, C. (2026). Inventory demand forecasting using Cat Boost regression: A machine learning method. In 2026 2nd International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI). IEEE. <https://doi.org/10.1109/IC3ECSBHI67834.2026.11468899>
- [17] Zhang, Z. (Justin), Srivastava, P. R., Eachempati, P., & Yu, Y. (2023). An intelligent framework for analyzing supply chain resilience of firms in China: A hybrid multicriteria approach. *International Journal of Logistics Management*, 34(2), 443-472. <https://doi.org/10.1108/IJLM-11-2020-0452>
- [18] Phu, B. Q., Phuc, P. H., & Son, P. T. (2026). Car price analysis using data collected from an online sales platform. *International Journal of Information Technology and Computer Science (IJITCS)*, 18(1), 124-139. <https://doi.org/10.5815/ijites.2026.01.07>
- [19] Awad, M., & Khanna, R. (2015). *Efficient learning machines: Theories, concepts, and applications for engineers and system designers*. Apress. <https://doi.org/10.1007/978-1-4302-5990-9>
- [20] Chinnnasamy, P. (2025). Transforming e-commerce with intelligent recommendation systems: A review of current trends in machine learning and deep learning. *International Journal of Computational and Experimental Science and Engineering*, 11(2), 1645-1661. <https://doi.org/10.22399/ijcesen.1183>
- [21] Fathalla, A., Salah, A., Li, K., & others. (2020). Deep end-to-end learning for price prediction of second-hand items. *Knowledge and Information Systems*, 62(12), 4541-4568. <https://doi.org/10.1007/s10115-020-01495-8>
- [22] Shukla, S., Maheshwari, A., & Johri, P. (2021). Comparative analysis of ML algorithms & Stream Lit web application. In *Proceedings of the 2021 3rd International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)* (pp. 175-180). IEEE. <https://doi.org/10.1109/ICAC3N53548.2021.9725496>
- [23] Zhou, F., Jiang, Y., Qian, Y., Liu, Y., & Chai, Y. (2024). Product consumptions meet reviews: Inferring consumer preferences by an explainable machine learning approach. *Decision Support Systems*, 177, 114088. <https://doi.org/10.1016/j.dss.2023.114088>
- [24] Kim, S., Shin, W., & Kim, H.-W. (2024). Predicting online customer purchase: The integration of customer characteristics and browsing patterns. *Decision Support Systems*, 177, 114105. <https://doi.org/10.1016/j.dss.2023.114105>
- [25] Tanvir, A.-A., Khandokar, I. A., Islam, A. K. M. M., Islam, S., & Shatabda, S. (2023). A gradient boosting classifier for purchase intention prediction of online shoppers. *Heliyon*, 9(4), e15163. <https://doi.org/10.1016/j.heliyon.2023.e15163>